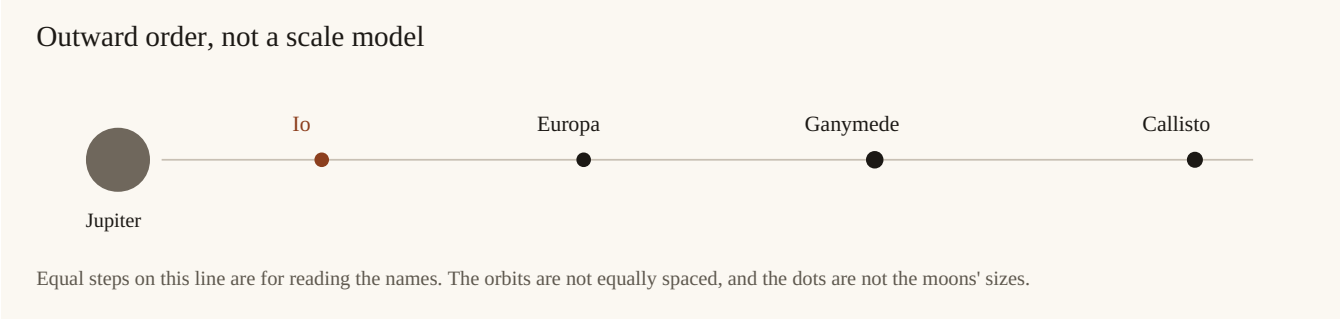
FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

Galileo's moons: new centers and lasting names

On 7 January 1610 Galileo saw points of light beside Jupiter, lined up east and west and changing place from night to night. He came to understand four bodies going around Jupiter. Sidereus Nuncius, in March 1610, told Europe. The fuller story of his telescope and his trial is the Galileo page. This page is only the moons, and the names.

They matter because not every moving body has to orbit the Earth. One old objection to Copernicus was that a traveling Earth would leave its Moon behind. Jupiter travels, in any of the systems then argued, and it keeps four companions. That weakens the objection. It does not finish the case. Tycho Brahe's system, Sun around the Earth and the other planets around the Sun, can hold moons around Jupiter just as well. New centers of motion are real. They are not, by themselves, a proof of a moving Earth.

Galileo named the four, together, the Medicean Stars, for the Medici who paid the patronage. He did not give them the names on a modern chart. Those names, Io, Europa, Ganymede, and Callisto, in that order outward, are Simon Marius's, in Mundus Iovialis of 1614. Marius said Kepler suggested them.



The four moons in outward order, not to scale

A reading order, drawn for this lesson. The gaps are equal so the names can be read. The orbits are not equally spaced, and the dots are not the moons' sizes.

Marius also said he first saw the moons on 29 December 1609. That date is Julian. On the Gregorian calendar it is 8 January 1610, the day after Galileo's 7 January. Do not treat the two calendars as if they were the same day, and do not call Marius a fraud because the numbers sit close. Who published a dated book first is one question. Who looked, independently, is another. The quarrel is real. A collapsed calendar is not evidence.

Io, the inner one, later became a clock. When it enters Jupiter's shadow the reflected sunlight stops, and the timetable of those eclipses taught Rømer that light takes time. That is the next lesson.

The illustrated lesson is download the PDF.

Sources

- Museo Galileo, on Jupiter's moons: https://catalogue.museogalileo.it/indepth/JupitersMoonsSatellites.html
- Museo Galileo, Sidereus Nuncius (1610): https://www2.museogalileo.it/it/biblioteca-e-istituto-di-ricerca/progetti/incontra-galileo/31-opere/454-sidereus-nuncius-1610.html
- Pasachoff, "Simon Marius's Mundus Iovialis and the discovery of the moons of Jupiter," IAU proceedings, 2018: https://ui.adsabs.harvard.edu/abs/2018iau3.book...38P/abstract

Key facts

- Galileo recorded moons of Jupiter from 7 January 1610 and published them in March 1610 as the Medicean Stars.
- Moons around Jupiter answer one objection to a moving Earth. Tycho's system can include them too. They are not a complete proof of Copernicus.
- Io, Europa, Ganymede, and Callisto are Marius's 1614 names, which he credited to a suggestion from Kepler.
- Marius's 29 December 1609 is Julian. Gregorian, it is 8 January 1610. Do not equate the calendars or assume fraud from the numbers.